15A, 600V Low Loss Single-Phase Single In-Line Bridge Rectifier

FEATURES

- Qualification reference AEC-Q101
- Ideal for printed circuit board
- Low forward voltage drop
- High surge current capability
- UL Recognized File # E-326243
- High case dielectric strength of 2000 V_{RMS}
- RoHS Compliant
- Halogen-free

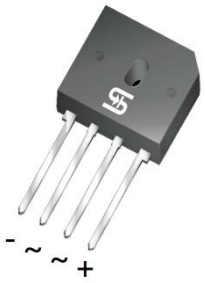
KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	15	A
V_{RRM}	600	V
I_{FSM}	280	A
$T_{J\ MAX}$	175	°C
Package	GBU	
Circuit Configuration	In-line	

APPLICATIONS

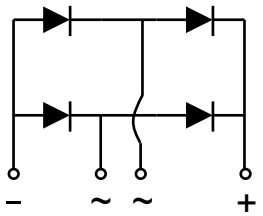
- Switching mode power supply (SMPS)
- AC to DC
- On board charger

MECHANICAL DATA

- Case: GBU
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: As marked
- Weight: 4.03g (approximately)



GBU



ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)				
PARAMETER		**SYMBOL**	**VALUE**	**UNIT**
Repetitive peak reverse voltage		V_{RRM}	600	V
Reverse voltage, total rms value		$V_{R(RMS)}$	420	V
Forward current		I_F	15	A
Surge peak forward current, single half sine-wave superimposed on rated load per diode	t = 8.3ms	I_{FSM}	280	A
	t = 1.0ms		620	
Rating of fusing (t<8.3ms)		I^2t	325	A^2s
Junction temperature		T_J	- 55 to +175	°C
Storage temperature		T_{STG}	- 55 to +175	°C

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-lead thermal resistance	$R_{\Theta JL}$	1.2	°C/W
Junction-to-ambient thermal resistance	$R_{\Theta JA}$	6.6	°C/W
Junction-to-case thermal resistance	$R_{\Theta JC}$	1.0	°C/W

Thermal Performance Note: Mounted on Heat sink with 4" x 6" x 0.25" Al-Plate

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER	CONDITIONS	SYMBOL	TYP	MAX	UNIT
Forward voltage per diode(1)	I_F = 7.5A, T_J = 25°C	V_F	0.87	0.9	V
	I_F = 7.5A, T_J = 125°C		0.76	-	V
Reverse current @ rated V_R per diode(2)	T_J = 25°C	I_R	-	10	μA
	T_J =125°C		0.5	-	μA
Junction capacitance per diode	1MHz, V_R = 4.0V	C_J	148	-	pF
Reverse recovery time	I_F = 0.5A, I_R = 1.0A, I_{rr} = 0.25A	t_{rr}	915	-	ns

Notes:

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

ORDERING INFORMATION

ORDERING CODE	PACKAGE	PACKING
GBUL15J	GBU	20 / Tube

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Forward Current Derating Curve

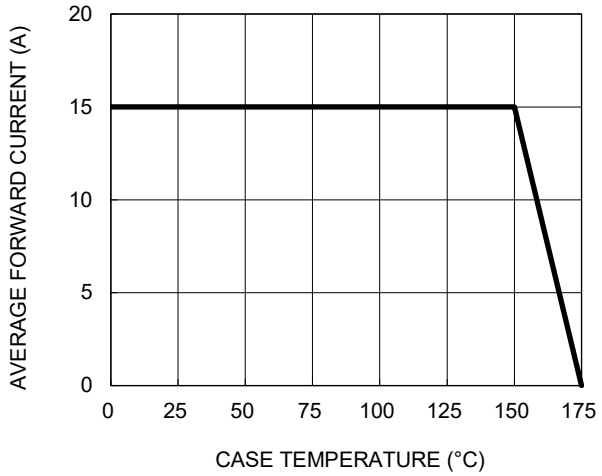


Fig.2 Typical Junction Capacitance

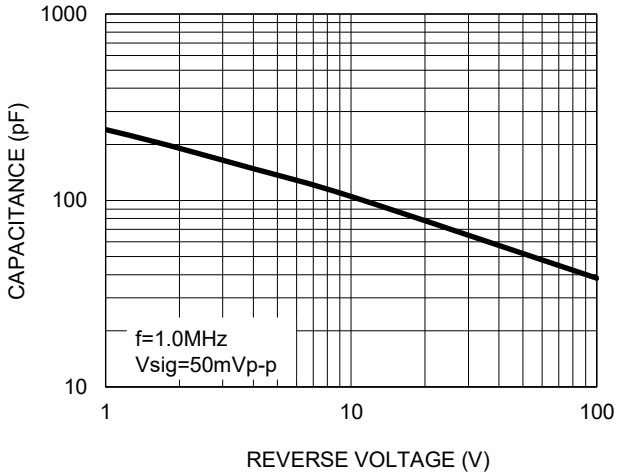


Fig.3 Typical Reverse Characteristics

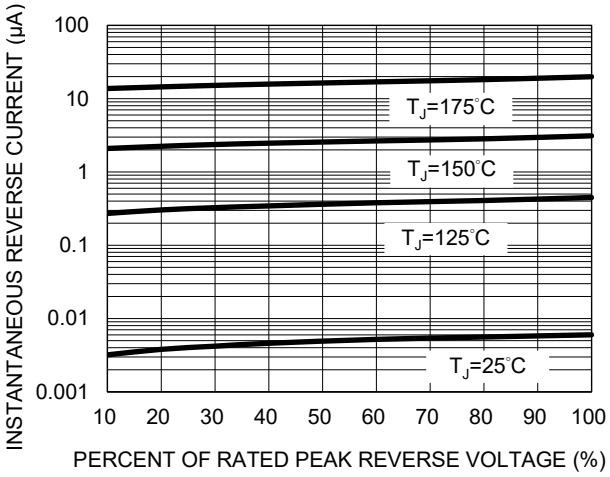


Fig.4 Typical Forward Characteristics

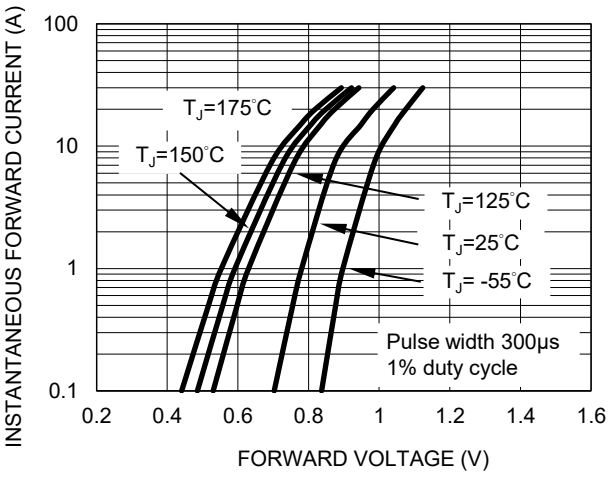
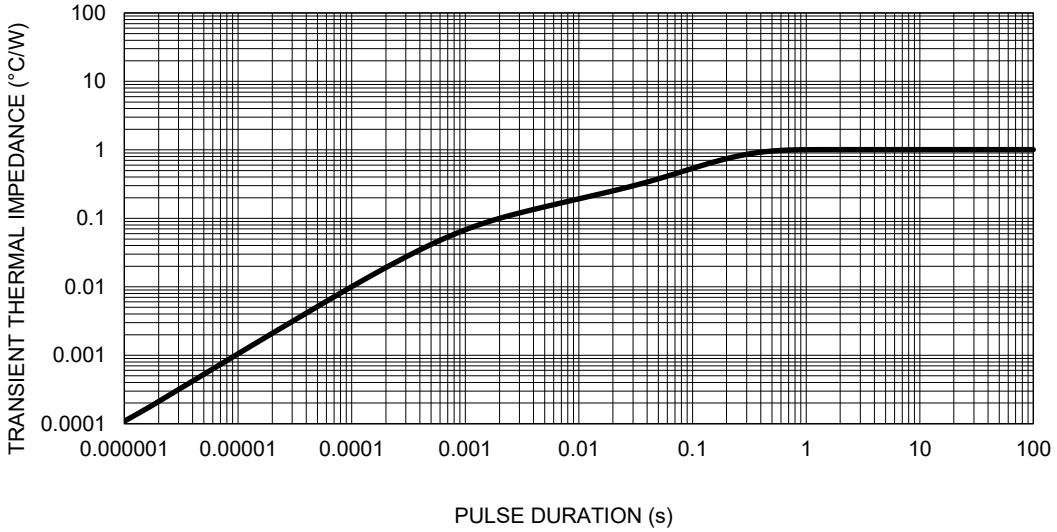
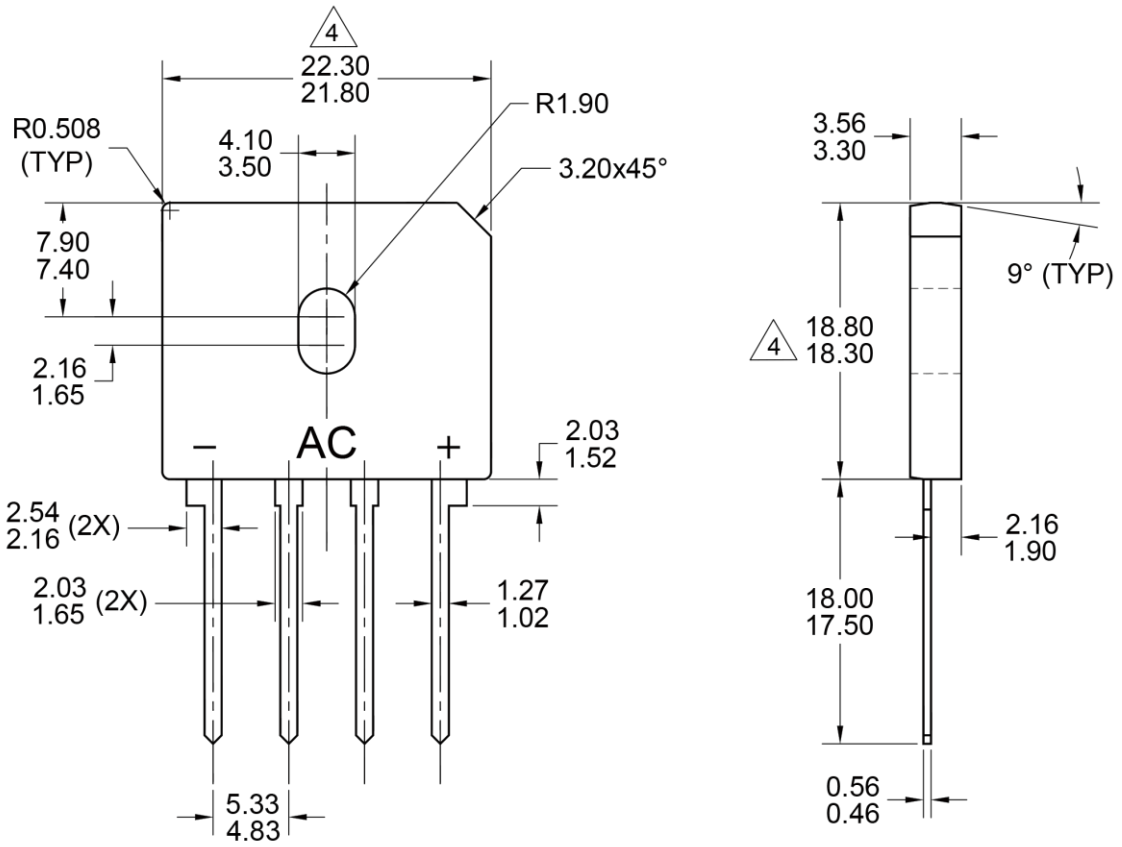


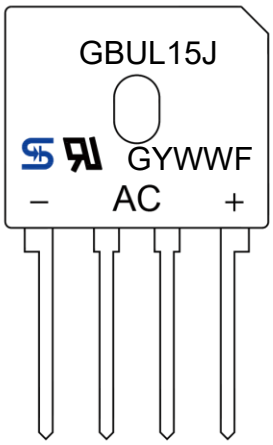
Fig.5 Typical Transient Thermal Impedance



PACKAGE OUTLINE DIMENSIONS

GBU





MARKING DIAGRAM

G = GREEN COMPOUND
YWW = DATE CODE
F = FACTORY CODE

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. THERE IS NO EXISTING PACKAGE OUTLINE INDUSTRY STANDARD FOR THIS PACKAGE.
4. MOLDED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
5. DWG NO. REF: HQ2SD07-GBU-068 REV A.

Notice

Specifications of the products displayed herein are subject to change without notice. TSC or anyone on its behalf, assumes no responsibility or liability for any errors or inaccuracies.

Purchasers are solely responsible for the choice, selection, and use of TSC products and TSC assumes no liability for application assistance or the design of Purchasers' products.

Information contained herein is intended to provide a product description only. No license, express or implied, to any intellectual property rights is granted by this document. Except as provided in TSC's terms and conditions of sale for such products, TSC assumes no liability whatsoever, and disclaims any express or implied warranty, relating to sale and/or use of TSC products including liability or warranties relating to fitness for a particular purpose, merchantability, or infringement of any patent, copyright, or other intellectual property right.

The products shown herein are not designed for use in medical, life-saving, or life-sustaining applications. Customers using or selling these products for use in such applications do so at their own risk and agree to fully indemnify TSC for any damages resulting from such improper use or sale.